

Super Information Theory: Coherence, Time Density, Informational Dissipation, and Gravity

A Correction-Integrated, Layered Research Program

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Abstract

Imagine two otherwise matched clocks or matter preparations with the same modeled mass-energy and geometry but different internal event or phase organization. General relativity assigns clocks and matter a spacetime metric; scalar-tensor models can add fields that alter that metric; and quantum information supplies preparation-dependent statistics. None of those inherited facts alone makes a measured event or coherence statistic a gravitational field. The open question is whether a reproducible residual remains after ordinary preparation energy, stress, electromagnetic, thermal, geometric, and mechanical effects are accounted for.

The candidate operation studied here is correspondingly conditional: declare event opportunities per unit proper time or a phase-alignment statistic, calibrate that measured quantity to a model field, let the field enter the metric followed by clocks and matter, and test whether reversing its spatial gradient reverses a residual. The event-rate description is called time density only after the apparatus and event class are declared. In a separate bridge, history-conditioned reduction of a declared mismatch is called informational dissipation only when the update rule, disturbance, physical substrate, and energy or entropy accounting are supplied. The full architecture is called Super Information Theory (SIT), and its gradient-to-motion subchain is called Quantum Gradient Time Crystal Dilation (QGTCD). A null tracking result excludes the tested coupling and preparation regime; it does not erase the historical question or every possible implementation.

For one frozen benchmark, each extra scalar mode contributes an exponentially range-suppressed correction to the ordinary inverse-distance potential - the standard form called a Yukawa correction. The temporal and spatial metric responses can then be compared by a local ratio; only in the appropriate long-range, fixed-mass convention does that ratio reduce to the conventional post-Newtonian parameter. Cassini radio tracking tests a fitted light-time model, while planetary orbits help calibrate the central mass, so a finite-range claim requires those observations and nuisance parameters to be fitted together.

Super Information Theory (SIT) is a layered research program connecting a historical sequence of ideas about coincidence, oscillatory organization, time density, informational dissipation, quantum state preparation, and gravity. Version 79 placed too many definitions, hypotheses, model choices, proposed derivations, comparisons, and applications inside one omnibus document. A later Version 80 correction-note draft correctly identified several local mathematical defects but then collapsed the wider program into a generic two-scalar model and treated that correction note as the whole successor. This paper takes a different route: it preserves the valid corrections while reconstructing the architecture they do not invalidate.

The paper makes eight contributions. First, it separates SIT into historical atoms, operational quantities, model variables, model-conditional derivations, empirical hypotheses, and companion research programs. Second, it specifies a consistent scalar-tensor baseline with two neutral real fields, an independent Maxwell sector, one physical matter metric, and an optional global phase sector that is not identified with electromagnetism. Third, it derives the baseline field equations and its weak-field observable potential. This produces a precise conditional version of the QGTCD mechanism: if a calibrated time-density field enters the physical matter metric, its spatial gradient contributes to acceleration through the gradient of that metric coupling. Fourth, it replaces the false claim that purity decreases under every quantum channel with operation-class-aware statements: purity monotonicity

is restricted to unital channels, while relative-entropy contraction is used only under declared coarse-graining maps. Fifth, it freezes and analytically solves one minimal scalar-tensor benchmark, exposing the exact Yukawa force law and stability conditions rather than leaving the central gradient mechanism at the level of an unspecified model family. Sixth, it translates that benchmark into standard post-Newtonian and Yukawa variables, derives a literature-contested long-range Cassini diagnostic proxy, identifies what the short-range torsion-balance result does and does not exclude, and separates direct null-sector cancellation from the distinct question of how a Keplerian mass calibration enters a fitted Shapiro coefficient. Seventh, it implements a deterministic synthetic paired-fit benchmark and an exact uniform-sphere finite-source response. That calculation shows when null and massive-body observations can identify two Yukawa modes, when equal or very long ranges destroy identifiability, and why one fixed apparatus geometry can hide a coupling bias even when residuals appear perfect. Eighth, it turns the published Cassini estimate into an executable, sign-constrained Gaussian summary-statistic likelihood and records the exact boundary between that calculation and a genuine Doppler, ephemeris, and nuisance-parameter reanalysis. A governed correction-intent ledger and public-source custody ledger accompany the manuscript so that mathematical repair does not erase the theory’s earlier questions or assign later wording to earlier sources.

SIT is not presented here as a proved unification of quantum mechanics and gravity. The baseline action is a conventional research scaffold; its distinctive role is to express and test a specific genealogy and coupling architecture. A microscopic state-to-field map, controlled effective-field-theory completion, full finite-source and multiscale constraint fit, and confirmed SIT-specific signal remain open. The proposed experimental program therefore targets residual differences between otherwise matched preparations rather than claiming fixed anomalies in advance.

Scope and evidence boundary

<https://selfawarenetworks.com> **Source lineage:** correction-integrated successor to Version 79, Zenodo record 18133332, concept DOI 10.5281/zenodo.16922511

1. Purpose and version relation

1.1 The physical problem in plain language

Suppose an apparatus prepares the same kind of clock, atom cloud, or material sample in two controlled internal organizations. Standard physics first requires a complete account of the energy, pressure, binding, fields, temperature, geometry, motion, and readout system used in those preparations. Only after that accounting is there a meaningful residual question.

The plain mechanism tested in this paper has four steps:

1. define what the apparatus counts or compares, such as resolved event opportunities or a declared phase-alignment statistic;
2. calibrate that measured statistic to a model field rather than identifying the two by name;
3. specify how the field changes the physical metric followed by clocks and matter; and
4. reverse or translate the resulting spatial gradient and test whether a residual clock shift or acceleration reverses with it.

General relativity supplies the inherited metric account, scalar-tensor theory supplies a conventional way to represent an added metric-coupled field, and quantum information supplies operation- and basis-dependent state comparisons. SIT names the joined research architecture only after those inherited pieces and the calibration step are kept distinct. QGTCD names the narrower field-gradient-to-motion chain. The decisive discriminator is not the name: it is a controlled, reversible residual that survives the standard account.

1.2 Version relation and correction rule

Version 79 remains the historical record of a broad attempt to integrate Super Dark Time, Quantum Gradient Time Crystal Dilation (QGTCD), SuperTimePosition, Micah’s New Law of Thermodynamics, and selected Self Aware Networks ideas into one physical framework. It also contains statements that should not be repeated as established results. The present draft follows two rules:

1. a false derivation is corrected at the smallest level that the counterexample reaches; and
2. a local correction does not erase a distinct historical atom, empirical question, or alternative model implementation.

The preserved Draft 6 correction note is therefore a companion technical ledger, not the complete replacement for SIT. Its counterexamples remain binding. Its editorial decision to reduce Version 80 to a correction-only paper does not.

The scientific posture of Draft 13 is:

SIT is a layered research program proposing that measurable event structure, phase organization, and history-dependent dissipation may admit effective fields whose coupling to physical clocks and matter can be tested. The paper supplies one consistent baseline realization and derives its conditional consequences. It does not claim that this realization is unique, ultraviolet complete, empirically confirmed, or already derived from microscopic quantum theory.

2. Public conceptual genealogy

SIT did not begin with Version 79’s action. Its architecture emerged through a sequence of public records. The dates establish provenance of the questions and conceptual combinations; they do not validate the physics.

2.1 Coincidence, oscillation, and information

The early program treated information as physically instantiated coincidence and relation rather than a detached symbol. Self Aware Networks later developed oscillatory coincidence, prediction error, phase difference, network-level rendering, and recurrent control. These neuroscience ideas supplied the conceptual bridge from static information measures to dynamical event patterns.

2.2 QGTCD in June 2022

The current verified public Git floor for QGTCD is earlier than the date printed in Version 79’s documentary appendix.

- GitHub commit [40c0b37f](#), dated 2022-06-10, records a joined discussion of oscillators, mass, density, scale, electromagnetic phase structure, spacetime curvature, and time dilation.
- GitHub commit [4c451f01](#), dated 2022-06-12, explicitly describes gradients of time structure around mass as changing the available directions or probabilities of particle motion.
- GitHub commit [4e0cflee](#), dated 2022-11-28, extends the program toward galactic rotation, the Tully-Fisher relation, MOND, and Hubble-tension questions.

The durable QGTCD atom is not any one speculative microscopic picture in those notes. It is the proposed operator chain:

local physical state or mass-energy organization

- > change in local temporal structure
- > spatial gradient in that temporal structure
- > biased motion or effective gravitational response.

Draft 13 asks how that chain can be represented in a covariant model without treating its historical metaphors as already-derived equations.

2.3 Private July 2023 genealogy route

A privately held July 5, 2023 recording preserves an intermediate plain-language presentation of the same proposed chain. It uses a movie / frame-rate analogy: a uniform sequence of frames becomes a three-dimensional region with locally different update density; mass or density is then proposed to change the local number of temporal opportunities, whose gradient biases directional motion. This is authorial-intent and explanation evidence only. It is not a public priority date, an independently validated result, a scientific reference, or authority to backdate any later equation or wording.

The governed route is reviews/DRAFT11-JO-RECORDER-QGTCD-GENEALOGY-ROUTE-20260725.md. Its machine transcript remains paraphrase-grade pending human listening and speaker attribution. The 2022 public Git commits above remain the public QGTCD floor, and no quotation from the private recording is load-bearing in this manuscript.

2.4 2024 public bridge articles

The 2024 SVGN series made the gravity / time-density program public in increasingly explicit language. The first-party SVGN export preserves the publication pages, while the Internet Archive independently supplies later public capture floors:

- *New Unified Field Theory: Quantum Gradient Time Crystal Dilation*, published 2024-01-29 and captured by the Internet Archive on 2024-03-02;
- *Quantum Gradient Time Crystal Dilation Breaks the Assumption that Gravity Equals Metric Curvature Alone*, published 2024-02-15 and captured on 2024-04-21;
- *Explain It to Me Like I Am Six: Quantum Gradient Time Crystal Dilation Theory (Part 3)*, published 2024-03-28 and captured on 2024-04-21;
- *Quantum Gravity's New Frontier: Time, Density, and Information*, published 2024-10-21 and captured on 2024-11-02; and
- *Dark Time Theory: A Conversation about the Core Ideas Forming a New Frontier in Physics*, published 2024-10-22 and captured on 2024-11-07.

Version 79 printed five different Wayback timestamps and truncated slugs that now return errors. Draft 13 does not repeat them. Publication dates come from the first-party exported pages; the later archive dates are the currently verified independent capture floors and must not be backdated to the publication dates.

These articles bridge the compact 2022 notes to the later formal papers. They are first-party genealogy and explanatory development, not independent scientific confirmation.

2.5 2025 synthesis

In plain terms, the later sequence asked three different questions: can a system reduce locally measured mismatch through dissipation; can locally counted temporal opportunities be operationalized; and can a state description retain more than one temporal placement or phase relation. The names Micah's New Law of Thermodynamics, Super Dark Time, and SuperTimePosition respectively label those roles. SIT then

attempted to place those elements in one covariant framework. Draft 13 retains that order. Later synthesis increases the number and specificity of claims; it does not backdate every later equation to the earliest note.

3. A layered architecture instead of an all-or-nothing theory

The main repair is organizational. SIT contains at least six different kinds of object:

Layer	Object	What can establish it	What it does not establish
G: genealogy	dated conceptual atom or narrative connection	public source and stable timestamp	mathematical correctness or empirical truth
O: operational	measurement protocol or observable summary	apparatus, estimator, resolution, uncertainty	existence of a new field
M: model	chosen fields, action, coupling, boundary conditions	explicit mathematical definition	uniqueness or physical reality
D: derived	consequence of the frozen model	valid derivation under stated assumptions	truth outside the model or parameter regime
H: hypothesis	proposed map from model variable to measured system	discriminator, controls, falsifier	confirmation before data
C: companion	separate domain translation or technical program	its own sources, methods, and review	automatic validation of core SIT

This prevents four recurring errors. A timestamp cannot become a theorem. An operational proxy cannot silently become a fundamental field. A definition cannot become a prediction. A successful companion calculation cannot validate every SIT bridge.

Two separate status vocabularies prevent another category error. A **source disposition** records what happened to a sentence inherited from an earlier version: retracted, replaced, retained-with-boundary, separated, or editorial-correction. A **current claim status** records what kind of statement remains in the present paper: definition, derived, computational, empirical, or open. Retraction of one old sentence therefore does not imply that every neighboring research question was retracted; conversely, preserving a question does not preserve a failed derivation. The correction-intent ledger accompanying this draft applies both labels to every governed correction ID.

4. Operational quantities and model variables

4.1 Event-rate or time-density observations

Let an observer or apparatus O , with four-velocity u^μ , count a declared class of distinguishable events inside a spatial and temporal resolution cell ϵ . For a point process, define the conditional event intensity along the apparatus worldline by

$$\rho_t^{(O,\epsilon)} = \lim_{\Delta\tilde{s} \rightarrow 0} \frac{\mathbb{E}[\Delta N_{O,\epsilon} | \mathcal{H}_O]}{\Delta\tilde{s}}. \quad (1)$$

Here $\mathcal{H}_O$ is the declared measurement history used to condition the intensity. This is an operational quantity. It depends on the event class, detector, threshold, resolution, and physical proper time $d\tilde{s}$. It is not automatically a universal Lorentz scalar independent of the protocol.

The baseline model uses a real scalar field τ with mass dimension one. A calibration map

$$\rho_t^{(O,\epsilon)} = \mathcal{R}_{O,\epsilon}(\tau, \partial\tau, \sigma, \dots) + \eta_{O,\epsilon} \quad (2)$$

relates τ to a measured event rate only after the response function $\mathcal{R}$ and noise term η are specified. Equation (2) is an open measurement contract, not a definition that guarantees the field exists.

4.2 Coherence observations

Different domains use different coherence measures: fringe visibility, off-diagonal density-matrix norms in a chosen basis, phase-locking statistics, mutual information, correlation length, condensate fraction, or spectral concentration. Let

$$C_M(x) = \mathcal{C}_M[\text{data in protocol } M] \quad (3)$$

denote one declared measure. The subscript is essential. Quantum coherence is basis- and operation-class-dependent; purity measures mixedness and is not identical to every coherence measure.

The baseline model uses a neutral real scalar σ , also with mass dimension one. A state-to-field hypothesis has the form

$$C_M(x) = \mathcal{K}_M[\sigma](x) + \eta_M(x), \quad (4)$$

where the kernel $\mathcal{K}$ contains coarse graining, support, resolution, and calibration. Until a particular $\mathcal{K}$ is supplied and tested, σ is a model amplitude, not purity, visibility, or a universal resource measure.

4.3 Phase and mismatch

Phase remains part of the SIT genealogy even though a phase gradient does not derive electromagnetism. For a declared pair of oscillatory processes, define

$$\Delta\theta_{ab}(x) = \theta_a(x) - \theta_b(x) \pmod{2\pi}. \quad (5)$$

The measured significance of $\Delta\theta$ depends on the oscillator definition and reference frame. The useful cross-domain atom is that changes in relative phase can encode mismatch, alignment, or transport. The physical implementation must be supplied separately in each domain.

4.4 What would earn a state-to-field map

Equations (2) and (4) are deliberately open contracts. To turn either contract into a testable map, a study must specify, before looking at the held-out residual:

1. the physical preparation and the ordinary energy, stress, temperature, charge, field, motion, and geometry variables that may change with it;
2. the apparatus, event class or coherence statistic, basis where applicable, resolution cell, uncertainty model, and data-reduction procedure;
3. the candidate kernel family, its dimensions, boundary conditions, training data, regularization, and identifiable parameter combinations;
4. at least one standard-physics null model and one nuisance-rich alternative that can explain preparation-correlated residuals without a new field;

5. a held-out or reversal test in which the inferred field gradient changes as preregistered while modeled preparation energy and geometry remain matched;
6. an out-of-sample prediction for a second location, scale, apparatus, or preparation; and
7. a failure rule: the map is rejected or narrowed if it does not transport, if ordinary covariates absorb the residual, or if the inferred field depends arbitrarily on resolution or estimator choice.

This checklist separates a learned calibration from a renamed measurement. A successful fit of $\mathcal{R}$ or $\mathcal{K}$ on the same data used to define the field would not establish a new physical degree of freedom. The field interpretation becomes informative only if it compresses multiple measurements, transfers across a declared domain, and predicts a reversible consequence that its ordinary-physics competitors miss.

5. Baseline covariant model

5.1 Fields and frames

Use four-dimensional natural units with signature $(-+++)$. Let $g_{\mu\nu}$ be an Einstein-frame metric, τ and σ neutral real scalar fields, A_μ the physical electromagnetic connection, $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, and χ the ordinary matter fields. Matter rods and clocks couple to one declared physical metric,

$$\tilde{g}_{\mu\nu} = \Omega^2(\tau, \sigma) g_{\mu\nu}, \quad \Omega(\tau, \sigma) > 0. \quad (6)$$

The electromagnetic connection and conformal factor are deliberately assigned different symbols.

5.2 Action

The correction-integrated baseline action is

$$S = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} R - \frac{Z_\tau}{2} \nabla_\mu \tau \nabla^\mu \tau - \frac{Z_\sigma}{2} \nabla_\mu \sigma \nabla^\mu \sigma - V(\tau, \sigma) - \frac{B_F(\tau, \sigma)}{4} F_{\mu\nu} F^{\mu\nu} \right] + S_m[\tilde{g}_{\mu\nu}, A_\mu, \chi]. \quad (7)$$

Here $Z_\tau > 0$, $Z_\sigma > 0$, and $B_F > 0$ are necessary local sign conditions. With $[\tau] = [\sigma] = [A_\mu] = 1$, $[R] = 2$, $[F_{\mu\nu}] = 2$, and dimensionless Ω , B_F , Z_τ , Z_σ , every displayed Lagrangian-density term has mass dimension four.

Equation (7) is a conventional scalar-tensor research scaffold with a scalar-dependent electromagnetic kinetic function. The scaffold itself is not the novelty claim. It is used because it can express and test the proposed coherence / time-density architecture without the Version 79 category errors.

5.3 A minimal coherence-time link

One model realization of the proposed link is

$$V(\tau, \sigma) = V_\tau(\tau) + V_\sigma(\sigma) + \frac{m_L^2}{2} (\tau - \kappa\sigma)^2. \quad (8)$$

The final term energetically favors τ approximately equal to $\kappa\sigma$ when its stiffness dominates the relevant gradients, other potential terms, and declared matter or gauge sources. Define $\delta_L = \tau - \kappa\sigma$. Rearranging the τ equation of motion gives

$$\delta_L = \frac{Z_\tau \square \tau - V'_\tau - \frac{1}{4} B_{F,\tau} F_{\mu\nu} F^{\mu\nu} + \alpha_\tau T_m}{m_L^2} = \mathcal{O}\left(\frac{\mathcal{Q}}{m_L^2}\right), \quad (9)$$

where $\mathcal{Q}$ denotes the scale of the displayed source numerator. Thus $\tau \simeq \kappa\sigma$ is an adiabatic approximation only when

$$|\delta_L| = \frac{|\mathcal{Q}|}{m_L^2} \ll \Phi_{\text{link}}, \quad (1)$$

for a declared nonzero characteristic link scale Φ_{link} . In a regime where it is an appropriate denominator, one possible choice is $\Phi_{\text{link}} = \max(|\tau|, |\kappa\sigma|)$; no relative-error claim is made by dividing through a field value at a zero crossing. This is a model choice. It does not derive the coherence-time relation from information theory, nor does it identify σ with a quantum resource measure.

5.4 Optional global phase sector

SIT's phase-current language can be represented without identifying the phase with physical electromagnetism. Add an optional compact scalar θ , with $K(\sigma) > 0$ in the regime being treated, through

$$S_\theta = -\frac{1}{2} \int d^4x \sqrt{-g} K(\sigma) \nabla_\mu \theta \nabla^\mu \theta. \quad (10)$$

If the action is invariant under a constant shift of θ , the Noether current is

$$J_\theta^\mu = K(\sigma) \nabla^\mu \theta, \quad \nabla_\mu J_\theta^\mu = 0. \quad (11)$$

on the θ equation of motion. This is a conserved current in the optional model sector. It is not a universal conservation law for every operational measure of coherence. A hidden local U(1) extension is possible only as a separate model with its own gauge boson, spectrum, mixing, matter couplings, and constraints.

The separation is physically consequential. If a complex field $\Psi = v e^{i\theta}$ with nonzero modulus v is minimally charged under a gauge connection, then its kinetic term contains a quadratic contribution proportional to $q^2 v^2 A_\mu A^\mu$. That is a vector-mass term after symmetry breaking, not ordinary massless Maxwell electromagnetism. The neutral σ baseline and the global phase sector in Equation (10) avoid silently giving the photon a mass. Any charged condensate must therefore be introduced as a separately identified hidden or broken gauge model and tested against its own spectrum and mixing constraints.

6. Model-conditional derivations

6.1 Stress-energy and Einstein equation

Equations (12)-(18) describe the baseline (7) without the optional phase sector (10). If that sector is activated, its stress tensor and its contribution to the σ equation must be added. For constant Z_τ and Z_σ , the scalar stress-energy tensor is

$$\begin{aligned} T_{\mu\nu}^{(\phi)} = & Z_\tau \left(\nabla_\mu \tau \nabla_\nu \tau - \frac{1}{2} g_{\mu\nu} (\nabla \tau)^2 \right) \\ & + Z_\sigma \left(\nabla_\mu \sigma \nabla_\nu \sigma - \frac{1}{2} g_{\mu\nu} (\nabla \sigma)^2 \right) - g_{\mu\nu} V. \end{aligned} \quad (12)$$

The electromagnetic contribution is

$$T_{\mu\nu}^{(F)} = B_F \left(F_{\mu\rho} F_{\nu}{}^{\rho} - \frac{1}{4} g_{\mu\nu} F_{\rho\lambda} F^{\rho\lambda} \right). \quad (13)$$

With the Einstein-frame matter tensor $T_{\mu\nu}^{(m)}$, metric variation gives

$$M_{\text{Pl}}^2 G_{\mu\nu} = T_{\mu\nu}^{(\phi)} + T_{\mu\nu}^{(F)} + T_{\mu\nu}^{(m)}. \quad (14)$$

Equation (14) shows that the new fields source curvature if they carry stress-energy. It does not show that ordinary gravity is uniquely derived from information.

6.2 Scalar equations

Define the Einstein-frame matter tensor by $T_m^{\mu\nu} = 2(-g)^{-1/2} \delta S_m / \delta g_{\mu\nu}$ and the matter-coupling slopes

$$\alpha_A(\tau, \sigma) = \frac{\partial \ln \Omega}{\partial \phi^A}, \quad \phi^A \in \{\tau, \sigma\}. \quad (15)$$

Let $T_m = g_{\mu\nu} T_m^{\mu\nu}$. Under the stated conventions, scalar variation yields

$$\begin{aligned} Z_A \square \phi^A &= V_{,A} + \frac{1}{4} B_{F,A} F_{\mu\nu} F^{\mu\nu} - \alpha_A T_m, \\ A &\in \{\tau, \sigma\} \quad \text{with no sum over } A. \end{aligned} \quad (16)$$

Cross-kinetic terms or a curved field-space metric would add connection terms and require a new derivation. Equation (16) is tied to the frozen baseline (7).

6.3 Electromagnetic and exchange equations

Define the matter current by $J_m^\nu = -(-g)^{-1/2} \delta S_m / \delta A_\nu$. Electromagnetic variation then gives

$$\nabla_\mu (B_F F^{\mu\nu}) = J_m^\nu. \quad (17)$$

The matter stress tensor is not separately conserved in the Einstein frame when the conformal factor varies. Its exchange equation is

$$\nabla_\mu T_m^{\mu\nu} = \alpha_A T_m \nabla^\nu \phi^A + F^\nu{}_\rho J_m^\rho. \quad (18)$$

Equation (18) sums over the repeated field label $A \in \{\tau, \sigma\}$; this differs from the explicitly unsummed component equation (16).

The scalar, electromagnetic, and matter exchanges cancel in the divergence of the total right-hand side of (14), as required by the Bianchi identity. Observable calculations must ultimately be expressed in the physical matter frame.

7. The QGTCD weak-field operator

The most important restoration is a precise version of the earlier QGTCD chain. Let the Einstein-frame weak-field metric satisfy

$$g_{00} = -(1 + 2\Phi_E), \quad |\Phi_E| \ll 1, \quad (19)$$

normalize the background conformal factor to $\Omega_0 = 1$, and expand it as

$$\delta \ln \Omega = \alpha_{\tau 0} \delta \tau + \alpha_{\sigma 0} \delta \sigma + \mathcal{O}(\delta \phi^2). \quad (20)$$

Here the subscript 0 means that the coupling slope (15) is evaluated on the chosen background.

Since $\tilde{g}_{00} = \Omega^2 g_{00}$, the physical weak-field potential read by matter is

$$\Phi_{\text{phys}} = \Phi_E + \delta \ln \Omega. \quad (21)$$

The nonrelativistic acceleration is therefore

$$\mathbf{a} = -\nabla \Phi_{\text{phys}} = -\nabla \Phi_E - \alpha_{\tau 0} \nabla \delta \tau - \alpha_{\sigma 0} \nabla \delta \sigma. \quad (22)$$

If the linking potential (8) produces the adiabatic relation (9), then

$$\mathbf{a} \simeq -\nabla \Phi_E - (\kappa \alpha_{\tau 0} + \alpha_{\sigma 0}) \nabla \delta \sigma. \quad (23)$$

Equations (21)-(23) express the QGTCD operator in a testable form:

calibrated temporal / coherence field change

- > change in the physical clock-and-matter metric
- > spatial potential gradient
- > conditional acceleration residual.

This result is derived from the model, not from nature. It survives only if the action is healthy, the background exists, the state-to-field map is operational, and a nonempty coupling region survives existing constraints.

In the same stationary weak-field regime, a signal emitted at e and received at r by static observers has the leading gravitational frequency shift

$$\frac{\nu_r}{\nu_e} - 1 \simeq \Phi_{\text{phys}}(e) - \Phi_{\text{phys}}(r) = -\Delta \Phi_E - \Delta \ln \Omega, \quad (24)$$

in units $c = 1$, with Δ defined as the receiver value minus the emitter value. A local-clock-rate convention relative to a common stationary coordinate uses the opposite sign, so an experiment must state which observable is reported. Transition-specific effects from B_F or other matter couplings must be added separately. A universal conformal shift and a change in an atomic transition are distinct observables and must not be double counted.

8. Information, dissipation, and the corrected thermodynamic bridge

Version 79 claimed that density-matrix purity decreases under every completely positive trace-preserving (CPTP) map. An amplitude-damping channel acting on the maximally mixed qubit increases purity, so that theorem is false. Purity non-increase for every state is restricted, in fixed dimension, to unital CPTP maps.

The counterexample can be checked directly. For damping probability $0 < p \leq 1$, apply the usual amplitude-damping Kraus operators $K_0 = |0\rangle\langle 0| + \sqrt{1-p}|1\rangle\langle 1|$ and $K_1 = \sqrt{p}|0\rangle\langle 1|$ to $\rho = I/2$. The output is $\text{diag}((1+p)/2, (1-p)/2)$, whose purity is $(1+p^2)/2 > 1/2$. A second check blocks the earlier equality claim: complete dephasing is nonunitary, yet every density matrix already diagonal in the dephasing basis is fixed and therefore retains exactly the same purity. These examples retract only the universal monotonicity and unitary-equality statements; they do not erase the narrower question of which declared open-system processes contract which declared mismatch or distinguishability measure.

The broader dissipation question can be posed without repeating that error. For two quantum states ρ and ω satisfying the usual support condition and a declared CPTP map Φ , quantum relative entropy obeys the data-processing inequality

$$D(\Phi(\rho) \parallel \Phi(\omega)) \leq D(\rho \parallel \omega). \quad (25)$$

Equation (25) is a contraction of distinguishability under the same coarse-graining map applied to both states. It is not a statement that every physical process destroys information or that every coherence measure decreases.

For a classical or computational state z_n , a mismatch functional $\mathcal{M}(z_n; y_n)$ is contractive only when the update rule and metric make it so:

$$\mathcal{M}(z_{n+1}; y_{n+1}) \leq q \mathcal{M}(z_n; y_n) + \varepsilon_n, \quad 0 \leq q < 1, \quad \varepsilon_n \geq 0. \quad (26)$$

The claim requires one common $q < 1$ on a declared state-target domain $\mathcal{D}$; it is not asserted globally outside that domain. The disturbance term ε_n includes exogenous disturbance and target drift $y_n \rightarrow y_{n+1}$, preventing the inequality from becoming a universal equilibrium claim in a changing environment. A thermodynamic interpretation additionally requires a physical substrate, temperature or reservoir model, work and heat accounting, and a declared entropy-production balance. SIT's retained hypothesis is narrower:

Some physical and computational systems may implement history-conditioned mismatch reduction through structured dissipation. The mapping from that contraction to thermodynamic entropy production must be derived for the specified system.

This statement connects SIT to the reconstructed MNLT paper without making MNLT a proof of the gravitational sector.

9. Experimental program as discriminators, not promised anomalies

The family action does not predict a number until its functions, parameters, background, boundary conditions, apparatus, and nuisance model are fixed. Draft 13 retains the experimental questions while withdrawing unsupported fixed effect sizes.

9.1 Matched coherence-preparation residual

Prepare two samples of the same species with matched total mass-energy, geometry, center-of-mass trajectory, electromagnetic environment, temperature budget, and known interaction energies, but different declared coherence statistic C_M . Define

$$\Delta a_{\text{res}} = (a_{\text{high-C}} - a_{\text{low-C}}) - \Delta a_{\text{standard}}. \quad (27)$$

The standard correction must include preparation energy, density profile, mean-field energy, magnetic sensitivity, vibration, wave-front aberration, thermal expansion, and analysis bias. SIT becomes relevant only if a preregistered residual tracks the declared state-to-field model and survives those controls.

9.2 Clock and interferometer test

For a frozen benchmark, calculate $\Delta \ln \Omega$ and any transition-specific shift from B_F . Compare co-located or path-matched clocks or interferometers under alternating preparations. The primary null is standard general relativity plus established atomic, electromagnetic, thermal, and mechanical systematics. A null result excludes only the tested coupling region.

9.3 Spatial-gradient test

Equation (22) predicts a gradient response, not an effect from a spatially uniform field value alone. A useful apparatus therefore reverses or translates the engineered gradient while keeping total preparation energy and other fields fixed. A signal that does not reverse with the modeled gradient is evidence against that benchmark.

9.4 Astrophysical and cosmological tests

Rotation curves, lensing, expansion history, structure growth, and clock data can constrain selected V, Ω, B_F functions. They do not support SIT merely because an extra scalar can fit data. Model comparison must penalize added flexibility and compare against dark matter, modified-gravity, baryonic, calibration, and selection-effect alternatives.

9.5 Falsification classes

A benchmark fails by at least one of four routes:

1. **internal failure:** the equations do not follow from the action, or no healthy background exists;
2. **interpretive failure:** the operational map (2) or (4) fails even if the scalar model is mathematically definable;
3. **constraint failure:** no parameter region survives laboratory, solar-system, astrophysical, cosmological, particle, or clock bounds; or
4. **experimental failure:** a preregistered discriminator excludes the remaining region.

10. Benchmark A: a solved linear reference model

The family model becomes scientifically sharper when at least one member is frozen far enough to calculate its force law. Benchmark A is deliberately simple and unscreened. It is not selected because it is expected to evade every existing bound. It is selected because it makes the QGTCD gradient operator, its degeneracies, and its failure conditions analytically transparent.

10.1 Frozen functions and background

Set $Z_\tau = Z_\sigma = 1$ and choose

$$\begin{aligned} V_A(\tau, \sigma) &= \frac{1}{2}m_\tau^2\tau^2 + \frac{1}{2}m_\sigma^2\sigma^2 + \frac{1}{2}m_L^2(\tau - \kappa\sigma)^2, \\ \Omega_A(\tau, \sigma) &= \exp\left(\frac{\beta_\tau\tau + \beta_\sigma\sigma}{M_{\text{Pl}}}\right), \quad B_F^{(A)} = 1, \end{aligned} \quad (28)$$

with $m_\tau^2 > 0$, $m_\sigma^2 > 0$, $m_L^2 > 0$, real $\kappa, \beta_\tau, \beta_\sigma$, and vanishing background fields in asymptotically Minkowski spacetime. The superscript (A) labels Benchmark A and is not the field derivative comma notation $B_{F,A}$ used in Equation (16). The constant electromagnetic kinetic function removes varying-coupling effects from this reference case. It does not remove the universal scalar force generated by Ω_A .

For $\phi = (\tau, \sigma)^T$, the quadratic mass matrix is

$$\mathbf{M}_A^2 = \begin{pmatrix} m_\tau^2 + m_L^2 & -\kappa m_L^2 \\ -\kappa m_L^2 & m_\sigma^2 + \kappa^2 m_L^2 \end{pmatrix}. \quad (29)$$

Its trace and determinant are

$$\begin{aligned} \text{tr } \mathbf{M}_A^2 &= m_\tau^2 + m_\sigma^2 + (1 + \kappa^2)m_L^2 > 0, \\ \det \mathbf{M}_A^2 &= m_\tau^2 m_\sigma^2 + \kappa^2 m_\tau^2 m_L^2 + m_\sigma^2 m_L^2 > 0. \end{aligned} \quad (30)$$

Therefore both eigenvalues are positive:

$$m_\pm^2 = \frac{1}{2} \left[\text{tr } \mathbf{M}_A^2 \pm \sqrt{(M_{\tau\tau}^2 - M_{\sigma\sigma}^2)^2 + 4\kappa^2 m_L^4} \right] > 0. \quad (31)$$

With canonical kinetic terms, the flat-background quadratic scalar sector has no ghost and has standard hyperbolic principal part. This is a local linear stability result only. It does not establish nonlinear, cosmological, radiative, or compact-object stability.

10.2 Point-source solution

Let $\mathbf{b} = (\beta_\tau, \beta_\sigma)^T$, and let the orthogonal matrix $\mathbf{U} = (\mathbf{u}_+, \mathbf{u}_-)$ diagonalize the mass matrix:

$$\mathbf{U}^T \mathbf{M}_A^2 \mathbf{U} = \text{diag}(m_+^2, m_-^2). \quad (32)$$

For nonrelativistic matter, $T_m \simeq -\rho$. To leading order around the frozen background, Equation (16) becomes

$$(\nabla^2 \mathbf{I} - \mathbf{M}_A^2) \phi = \frac{\mathbf{b}}{M_{\text{Pl}}} \rho. \quad (33)$$

For a point source of mass M , $\rho(\mathbf{x}) = M\delta^3(\mathbf{x})$, the solution that vanishes at infinity is

$$\phi(r) = -\frac{M}{4\pi M_{\text{Pl}} r} \mathbf{U} \begin{pmatrix} e^{-m_+ r} & 0 \\ 0 & e^{-m_- r} \end{pmatrix} \mathbf{U}^T \mathbf{b}. \quad (34)$$

Using the bare Einstein-frame coupling $G_* \equiv (8\pi M_{\text{Pl}}^2)^{-1}$, the physical potential is

$$\Phi_{\text{phys}}(r) = -\frac{G_* M}{r} \left[1 + 2 \sum_{i \in \{+, -\}} (\mathbf{b} \cdot \mathbf{u}_i)^2 e^{-m_i r} \right]. \quad (35)$$

The radial acceleration of a universal test body is consequently

$$\mathbf{a}(r) = -\frac{G_* M}{r^2} \left[1 + 2 \sum_{i \in \{+, -\}} (\mathbf{b} \cdot \mathbf{u}_i)^2 (1 + m_i r) e^{-m_i r} \right] \hat{\mathbf{r}}. \quad (36)$$

Equations (35)-(36) are the explicit prediction of Benchmark A's linear static sector. They are a two-mode Yukawa correction to the Newtonian potential. That functional form is standard scalar-tensor physics; it is not by itself an SIT-specific discovery. SIT-specific content would require a validated operational map from event rate or coherence preparation into τ, σ , followed by a residual that tracks that map.

10.3 What this benchmark clarifies

When m_L^2 dominates, the heavy mode penalizes $\tau - \kappa\sigma$, and the surviving light direction approaches $(\tau, \sigma) \propto (\kappa, 1)$. Its matter coupling is proportional to $\kappa\beta_\tau + \beta_\sigma$, reproducing the coupling combination in Equation (23). The historical QGTCD chain is therefore present in this benchmark as a conditional fifth-force operator.

The same calculation exposes the burden. Because Benchmark A has universal conformal coupling and no screening mechanism, long-range order-one values of β_τ or β_σ are disfavored by the diagnostic constant-coupling proxy derived below, but an experimental exclusion requires the real-data joint fit defined in Sections 10.5-10.6. The dependence on scalar masses and on the projection of the matter-coupling vector onto the mass eigenvectors is the standard multiscalar-tensor structure analyzed by Hohmann et al. [21]. Massless or effectively constant-coupling post-Newtonian bounds and range-dependent fifth-force limits are candidate tests of this unscreened benchmark. Its exactly universal test-body coupling does not by itself predict a composition-dependent weak-equivalence-principle violation. MICROSCOPE [20] becomes a direct discriminator only if a later extension introduces composition-dependent scalar charge through species-dependent kernels, electromagnetic couplings, binding-energy sensitivities, or another nonuniversal sector. Viability may require small projected couplings, short Compton wavelengths, a separately derived screening mechanism, or retirement of a proposed parameter subcase. Section 10.4 supplies an analytic diagnostic envelope; Sections 10.5-10.6 separate the experiment-level obligation from the bounded synthetic calculation now completed.

Benchmark A also separates two claims that Version 79 blurred:

1. **model result:** a scalar coupled through the physical matter metric generates a calculable gradient force; and
2. **SIT interpretation:** a measured time-density or coherence statistic maps to that scalar.

The first is now explicit. The second remains the central uncalibrated hypothesis.

10.4 Constraint map in the benchmark's own variables

The force law can now be translated without inventing a new parameter convention. Define the projected matter couplings, Compton ranges, and positive Yukawa sum by

$$\begin{aligned}
c_i &= \mathbf{b} \cdot \mathbf{u}_i, \\
\alpha_i &= 2c_i^2 \geq 0, \\
\lambda_i &= m_i^{-1}, \\
\mathcal{A}(r) &= \sum_{i \in \{+, -\}} \alpha_i e^{-r/\lambda_i}.
\end{aligned} \tag{37}$$

Here $c = \hbar = 1$ except where units are restored below. Equation (35) is therefore the standard positive-strength two-Yukawa potential. Orthogonality of $\mathbf{U}$ gives $\sum_i c_i^2 = \|\mathbf{b}\|^2 = \beta_\tau^2 + \beta_\sigma^2$; rotating the fields cannot hide total long-range coupling strength.

At the same linear point-source order, the physical matter-frame metric has temporal and spatial potentials proportional to $1 + \mathcal{A}(r)$ and $1 - \mathcal{A}(r)$, respectively. Before any post-Newtonian limit is invoked, define their local metric-potential ratio by

$$\begin{aligned}
\gamma_{\text{met}}(r) &= \frac{1 - \mathcal{A}(r)}{1 + \mathcal{A}(r)}, \\
\gamma_{\text{met}}(r) - 1 &= -\frac{2\mathcal{A}(r)}{1 + \mathcal{A}(r)} \leq 0.
\end{aligned} \tag{38}$$

At finite range, $\gamma_{\text{met}}(r)$ is first a ratio of local metric potentials, not a coefficient that may be inserted directly into a constant-parameter PPN likelihood. It reduces to the conventional constant-coupling PPN γ only in the appropriate long-range regime and after the gravitational mass convention is fixed.

The opposite scalar signs also impose a useful photon-sector identity. If $U_* = G_* M/r$ denotes the bare tensor potential, then $U_t = U_*[1 + \mathcal{A}(r)]$, $U_s = U_*[1 - \mathcal{A}(r)]$, and

$$[1 + \gamma_{\text{met}}(r)]U_t(r) = 2U_*(r). \tag{39}$$

Thus the direct conformal scalar contribution cancels from the leading null-propagation and lensing combination $U_t + U_s$. Any Cassini sensitivity can arise only through comparison with the solar mass parameter inferred from massive-body dynamics, whose acceleration contains the range-dependent factor in Equation (36). A dedicated single-scalar analysis argues that treating the published constant-PPN Cassini value as a direct constraint on $\gamma_{\text{met}}(r)$ is not rigorous at finite range because the modeled null observable loses the Yukawa terms [24]. For this two-mode benchmark, that result is a warning rather than a completed multiscalar fit: a finite-range analysis must fit ephemerides, mass calibration, null propagation, and nuisance parameters together, and it must not add a standalone scalar Shapiro delay to a fixed general-relativistic solar mass.

The factor of two follows from normalization rather than convention-shopping. In the dimensionless Damour–Esposito-Farese variables $\varphi_i = \phi_i/(\sqrt{2}M_{\text{Pl}})$, the standard matter-coupling derivative is $\alpha_i^{\text{DEF}} = \partial \ln \Omega / \partial \varphi_i = \sqrt{2}c_i$ [17]. The usual massless result $\gamma - 1 = -2\alpha_{\text{DEF}}^2/(1 + \alpha_{\text{DEF}}^2)$ therefore becomes the long-range PPN limit of Equation (38), while the finite-range metric ratio weights each eigenmode by its Yukawa factor. This is the Einstein-frame version of the mass-eigenvector and coupling-alignment dependence derived for general multiscalar-tensor gravity by Hohmann et al. [21]. It also fixes the sign: the healthy canonical Benchmark A predicts $\gamma_{\text{met}}(r) \leq 1$, not an arbitrary two-sided deviation.

Bertotti, Iess, and Tortora measured a constant-PPN fit $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ with the Cassini radio link [22]. For a transparent two-standard-deviation long-range diagnostic proxy, a nonpositive Benchmark

A deviation must remain above the lower value -2.5×10^{-5} . Writing $\delta_C = 2.5 \times 10^{-5}$, the long-range reading of Equation (38) gives

$$\mathcal{A}_C \leq \frac{\delta_C}{2 - \delta_C} = 1.2500156 \times 10^{-5}. \quad (40)$$

For modes whose Compton ranges are effectively constant across both the orbital mass-calibration geometry and the radio-propagation geometry, the exponentials approach one and the diagnostic becomes the invariant coupling proxy

$$\begin{aligned} \beta_\tau^2 + \beta_\sigma^2 &\leq \frac{\delta_C}{2(2 - \delta_C)} = 6.2500781 \times 10^{-6}, \\ \|\mathbf{b}\| &\leq 2.5000156 \times 10^{-3}. \end{aligned} \quad (41)$$

Equation (41) is a provisional long-range constant-coupling diagnostic, not a reanalysis of the Cassini Doppler data and not yet an experimental exclusion. It applies only when the Yukawa factors are effectively constant across both the solar-mass calibration and radio experiment. For finite ranges, replacing the joint likelihood by $\gamma_{\text{met}}(r)$ at one arbitrarily chosen distance is insufficient. A final constraint must recompute massive-body ephemerides, the inferred solar mass, null propagation, and the experiment's nuisance parameters in one model, and must explicitly resolve the contrary Cassini-insensitivity analysis [24].

At short range, Lee et al. tested the inverse-square law from $52 \mu\text{m}$ to 3.0 mm and reported at 95 percent confidence that a gravitational-strength Yukawa term, $\alpha = 1$, must have range below $38.6 \mu\text{m}$ [23]. Under Equation (37), $\alpha_i = 1$ means $|c_i| = 1/\sqrt{2}$, so this published anchor maps to

$$\begin{aligned} |c_i| = \frac{1}{\sqrt{2}} &\implies \lambda_i < 38.6 \mu\text{m}, \\ m_i c^2 &> \frac{\hbar c}{38.6 \mu\text{m}} = 5.11 \text{ meV}. \end{aligned} \quad (42)$$

The full two-mode exclusion cannot be read from this one quoted contour. The torsion balance responds to extended source and detector geometries, and its published single-Yukawa likelihood must be recomputed for the sum in Equation (37). Positivity prevents cancellation between the two canonical modes, but it does not justify applying a point-particle value at one separation to the entire apparatus. Section 10.6 solves a uniform-sphere finite-source example to expose the geometry dependence; that example is not the Lee et al. apparatus and does not replace its likelihood.

10.5 The mass-calibration dispute and the observable bracket

The cancellation in Equation (39) answers a narrow question: at this order, the bare tensor mass parameter $G_* M$ controls the direct null-sector integral because the opposite conformal scalar contributions to $U_t + U_s$ cancel. It does not by itself answer which mass parameter an observational pipeline holds fixed. That distinction is the center of a real literature dispute. Huang and Deng model the null observable with the bare coefficient and conclude that the Yukawa parameters disappear from Cassini tracking [24]. Dyadina and Labazova argue that the mass entering a Shapiro prediction is not directly measured; replacing it by a mass inferred from massive-body orbital dynamics - often called a Keplerian mass calibration - restores model dependence [25]. Planetary-ephemeris methodology independently warns that an alternative theory can

modify the coordinate framework, massive-body dynamics, light propagation, and fitted constants together [26].

For Benchmark A, the force law in Equation (36) makes that calibration question explicit. Define the two-mode massive-body response at a declared calibration radius r_K by

$$\mathcal{K}(r_K) = \sum_{i \in \{+, -\}} \alpha_i \left(1 + \frac{r_K}{\lambda_i} \right) e^{-r_K/\lambda_i}. \quad (43)$$

In the point-source, nearly circular, single-radius idealization, the mass parameter inferred from radial acceleration is

$$\mu_K(r_K) \equiv r_K^2 |a(r_K)| = G_* M [1 + \mathcal{K}(r_K)]. \quad (44)$$

If a GR-shaped Shapiro coefficient is then written as $[1 + \gamma_{\text{cal}}(r_K)]\mu_K$, while the direct null integral retains the coefficient $2G_*M$, the calibration identity is

$$\begin{aligned} 1 + \gamma_{\text{cal}}(r_K) &= \frac{2G_*M}{\mu_K(r_K)} = \frac{2}{1 + \mathcal{K}(r_K)}, \\ \gamma_{\text{cal}}(r_K) - 1 &= -\frac{2\mathcal{K}(r_K)}{1 + \mathcal{K}(r_K)}. \end{aligned} \quad (45)$$

Equation (45) is not a Cassini likelihood and is not a new PPN parameter. It is a bookkeeping identity that shows why the two published conclusions need not be verbal opposites: the direct null integrand can be scalar-free at leading order, while a coefficient expressed relative to a dynamically inferred mass can remain scalar-dependent. In the strictly long-range limit, $\mathcal{K} = \mathcal{A}$, and Equation (45) reduces to the diagnostic form of Equation (38). At finite range, $\mathcal{K}$ contains the force factor $1 + r_K/\lambda_i$ and cannot be replaced by $\mathcal{A}$ evaluated at the radio ray's impact parameter.

A real ephemeris does not infer one solar mass at one radius. It fits many trajectories, initial conditions, bodies, clocks, range and Doppler observations, nuisance parameters, and a common dynamical model. Therefore r_K in Equations (43)-(45) is an explanatory reduction, not the missing experiment-level analysis. The experiment-level discriminating computation is a paired fit:

1. a **bare-coefficient fit**, implementing the null cancellation with G_*M as the fitted common coefficient; and
2. a **dynamically calibrated fit**, estimating the mass and Yukawa parameters jointly from the same massive-body and radio data.

The difference in fit quality and posterior structure - not a borrowed constant-PPN value applied directly to $\gamma_{\text{met}}(r)$ - decides whether Cassini contributes information for this benchmark. Section 10.6 implements only the algebraic identifiability skeleton of that comparison using declared synthetic data. It does not contain the Cassini design matrix, covariance, clocks, trajectories, coordinate realization, or nuisance parameters.

The same benchmark has a more direct timelike target. To first order in the positive Yukawa strengths and in the nearly Keplerian approximation used in the cited single-field analysis, the two-mode anomalous periastron rate is additive:

$$\begin{aligned} \left. \frac{d\varpi}{dt} \right|_{\text{Yukawa}} &\simeq \frac{n}{2} \sum_{i \in \{+, -\}} \alpha_i \left(\frac{p}{\lambda_i} \right)^2 e^{-p/\lambda_i}, \\ p &= a(1 - e^2). \end{aligned} \tag{46}$$

The two-mode sum in Equation (46) inherits the cited single-mode expression only at first order for independent linear eigenmodes on the frozen background. Cross-mode nonlinearities, finite source structure, relativistic corrections beyond the stated approximation, and simultaneous parameter refitting are not contained in that formula.

Equation (46) supplies a direct route to planetary or binary timing, but it is still a perturbative point-source formula. Residual perihelion rates from a GR-built ephemeris are not a substitute for a global refit with the Yukawa dynamics included. Draft 13 therefore promotes the ephemeris / global-solve route above the Cassini proxy without pretending that such a fit has already been performed.

These results answer one limited algebraic proxy question. The family has a nonempty algebraic region under the diagnostic long-range proxy: the ball $0 < \|\mathbf{b}\| < 2.5 \times 10^{-3}$ is nonzero, while sufficiently short ranges suppress solar-system effects. This is not yet a Cassini-derived allowed or viable physical region and does not establish detectability, cosmological consistency, joint experimental admissibility, or an SIT-specific signal. Because SIT currently imposes no lower bound on $\|\mathbf{b}\|$, arbitrarily weak coupling always survives; any future claim requiring order-one coupling must instead clear the range-dependent laboratory bounds or derive a screening mechanism.

No separate leading clock-composition bound appears in frozen Benchmark A. The model sets $B_F = 1$ and couples ordinary matter universally through one physical metric, so standard clock redshift follows that same metric potential. A transition-specific clock signal, varying constant, or composition-dependent MICROSCOPE signal requires a nonuniversal extension and its own source and detector charges. Such extensions must not be counted as constraints on the present benchmark before they are defined.

10.6 What a paired fit must distinguish

The practical problem can be stated without new terminology. Suppose the same central source is measured in two ways. Massive bodies respond to the tensor contribution plus the range-dependent scalar force. At the order frozen here, a radio signal responds directly to the tensor coefficient because the scalar terms cancel from the leading null combination. The two observations can therefore share one underlying mass parameter while depending differently on the scalar modes. A useful analysis must estimate the shared quantity and the mode-dependent quantities together. If it uses only the radio coefficient, samples only one orbital radius, or gives both modes the same range, it cannot recover both mode strengths.

The source and detector are also not mathematical points. For a homogeneous sphere of radius R , the mass-normalized form factor multiplying its outside Yukawa potential is [27]

$$\begin{aligned} \mathcal{F}(x) &= \frac{3[x \cosh x - \sinh x]}{x^3}, \\ x &= \frac{R}{\lambda}, \\ \mathcal{F}(0) &= 1. \end{aligned} \tag{47}$$

For two nonoverlapping homogeneous spheres with source radius R_s , detector radius R_d , and center separation $r > R_s + R_d$, the force kernel of mode i becomes

$$f_i^{(g)}(r) = \mathcal{F}\left(\frac{R_s^{(g)}}{\lambda_i}\right) \mathcal{F}\left(\frac{R_d^{(g)}}{\lambda_i}\right) \left(1 + \frac{r}{\lambda_i}\right) e^{-r/\lambda_i}, \quad (48)$$

where g labels the apparatus geometry. Equation (48) is exact for the declared uniform, spherical, nonoverlapping case. It is not the geometry of a torsion balance, a layered Earth, or a planetary ephemeris.

At fixed ranges, define $q_i = \mu_* \alpha_i$, where $\mu_* = G_* M$. A normalized massive-body coefficient measured at radius r_j and geometry g_j has the linear form

$$y_j^{\text{dyn}} = \mu_* + \sum_{i \in \{+, -\}} q_i f_i^{(g_j)}(r_j) + \epsilon_j, \quad (49)$$

while a normalized leading null-sector coefficient has

$$y_k^{\text{null}} = 2\mu_* + \nu_k. \quad (50)$$

Equations (49)-(50) explain the operation before naming it: several dynamic radii supply differently shaped mode columns, and the null observations anchor the shared tensor coefficient. Stack the observations as

$$\begin{aligned} \mathbf{y} &= \mathbf{X}(\lambda_+, \lambda_-) \boldsymbol{\eta} + \boldsymbol{\epsilon}, \\ \boldsymbol{\eta} &= (\mu_*, q_+, q_-)^T. \end{aligned} \quad (51)$$

For declared covariance $\mathbf{W}^{-1}$, the weighted least-squares solution can be written with the Moore-Penrose pseudoinverse,

$$\hat{\boldsymbol{\eta}} = (\mathbf{W}^{1/2} \mathbf{X})^+ \mathbf{W}^{1/2} \mathbf{y}, \quad (52)$$

and, when $\hat{\mu}_* \neq 0$, the physical strengths are recovered by

$$\hat{\alpha}_i = \frac{\hat{q}_i}{\hat{\mu}_*}. \quad (53)$$

This joined operation is the **paired-fit identifiability benchmark** used here. The name does not turn the calculation into an ephemeris or radio-science analysis; it labels the simpler inverse problem that those analyses must contain.

10.6.1 Deterministic synthetic result

The executable benchmark freezes $\mu_* = 1.3$, $(\alpha_+, \alpha_-) = (0.012, 0.004)$, and $(\lambda_+, \lambda_-) = (0.35, 1.4)$ in arbitrary common units. It creates six dynamic observations in each of two spherical geometries, four null-coefficient observations, declared independent standard deviations, and a fixed small noise vector. No experimental data, randomized sampling distribution, p-value, or confidence claim enters this test.

Fit	Result
null coefficient alone	$\hat{\mu}_* = 1.30000006$; the full three-column design has rank 1, so neither scalar strength is identified

Fit	Result
joint finite-source model	$\hat{\mu}_* = 1.30000004$, $\hat{\alpha}_+ = 0.01199894$, $\hat{\alpha}_- = 0.00400071$; rank 3; $\chi^2 = 0.6276$ for 13 residual degrees of freedom
joint point-geometry model	$\chi^2 = 1482.90$ for 13 residual degrees of freedom under the deliberately precise synthetic uncertainties
no-Yukawa model	$\chi^2 = 3.2125 \times 10^6$ for 15 residual degrees of freedom

Those numerical gaps measure only recovery of the synthetic generator. They do not estimate experimental sensitivity or evidence for SIT.

The most informative result came from a negative control. For one fixed pair of spherical bodies, each $\mathcal{F}(R_s/\lambda_i)\mathcal{F}(R_d/\lambda_i)$ is only a constant rescaling of mode i . A point-source fit can absorb that factor into α_i and achieve the same residuals while returning biased couplings. In the exact single-geometry control, both fits have numerical zero residual, but the point model returns (0.0130574, 0.00402135) instead of (0.012, 0.004). Multiple body geometries break that equivalence only if they share the same underlying couplings and their geometry is known.

10.6.2 Failure regions

The deterministic rank and conditioning checks expose five boundaries:

1. null observations alone identify only μ_* ;
2. equal scalar ranges make the two mode columns identical, so only their sum is identifiable;
3. one dynamic radius plus null observations cannot separate two mode kernels;
4. ranges far longer than the sampled radii make the columns nearly collinear - the declared long-range test has condition number 1.97×10^{16} ; and
5. one fixed extended-body geometry cannot reveal a form-factor rescaling when the mode strengths are free.

The bounded calculation therefore closes an implementation gap without closing the empirical one. It proves that the paired design can recover its own two-mode synthetic generator under favorable rank and geometry conditions. It also proves that perfect residuals need not imply correctly inferred couplings. A real result still requires the actual equations of motion, light-time model, coordinate conventions, clocks, covariance, nuisance parameters, source and detector density profiles, selection rules, and blind validation appropriate to the data.

10.7 Executable likelihood from the published Cassini summary

The published Cassini result contains one real-data summary that can be used without pretending to possess the Doppler time series. Let $d_C = 2.1 \times 10^{-5}$ and $s_C = 2.3 \times 10^{-5}$ denote the reported central estimate and standard uncertainty for $\gamma - 1$ [22]. In the strictly long-range, constant-coupling interpretation of Equation (38), set $\mathcal{A}_0 = \sum_i \alpha_i = 2\|\mathbf{b}\|^2$ and

$$\begin{aligned} \delta\gamma(\mathcal{A}_0) &= -\frac{2\mathcal{A}_0}{1 + \mathcal{A}_0}, \\ \chi_{\text{sum}}^2(\mathcal{A}_0) &= \left[\frac{d_C - \delta\gamma(\mathcal{A}_0)}{s_C} \right]^2, \quad \mathcal{A}_0 \geq 0. \end{aligned} \tag{54}$$

This is a Gaussian likelihood for the published one-number estimate, not a likelihood for the radio observations. Because the measured central value is positive while this benchmark permits only $\delta\gamma \leq 0$,

the constrained maximum-likelihood estimate is $\hat{\mathcal{A}}_0 = 0$. Relative to that boundary point,

$$\begin{aligned}\Delta\chi_{\text{sum}}^2(\mathcal{A}_0) &= \frac{[d_C + x(\mathcal{A}_0)]^2 - d_C^2}{s_C^2}, \\ x(\mathcal{A}_0) &= \frac{2\mathcal{A}_0}{1 + \mathcal{A}_0} \geq 0.\end{aligned}\tag{55}$$

Using the conventional one-parameter, one-sided asymptotic threshold $\Delta\chi^2 = 2.71$ gives

$$\begin{aligned}x_{95} &= -d_C + \sqrt{d_C^2 + 2.71 s_C^2} = 2.22965 \times 10^{-5}, \\ \mathcal{A}_{0,95} &= \frac{x_{95}}{2 - x_{95}} = 1.11484 \times 10^{-5}, \\ \|\mathbf{b}\|_{95} &= \sqrt{\frac{\mathcal{A}_{0,95}}{2}} = 2.36097 \times 10^{-3}.\end{aligned}\tag{56}$$

Equation (56) is a sign-constrained published-summary proxy. It improves on the earlier two-standard-deviation arithmetic by using the reported central value, stated uncertainty, model sign, and an explicit likelihood-ratio rule. It still is not an exact confidence statement for Benchmark A. The boundary, the asymptotic threshold, correlations with the fitted solar mass and orbital parameters, and the contrary null-sector interpretation in References 24-26 all matter. If the proper joint light-time and ephemeris fit makes Cassini insensitive to the scalar modes, Equation (56) must not be advertised as a Cassini exclusion. Its value here is executable bookkeeping: it states exactly what follows if, and only if, the published constant-PPN summary is accepted as the long-range observable for this benchmark.

The short-range finite-source route cannot yet be upgraded in the same way. Lee et al. publish a single-Yukawa exclusion and describe the harmonic torque apparatus [23], but Draft 13 does not contain the experiment's full geometry implementation, likelihood or covariance, systematic-error model, and two-mode torque transfer. The uniform-sphere calculation therefore remains a geometry and identifiability demonstration, not a laboratory constraint fit.

11. Corrections retained from Version 79

Correction ID	Version 79 statement	Draft 13 source disposition	What remains open
SIT79-H-001	purity decreases under every CPTP map	retracted; retain the unital-channel restriction and relative-entropy data processing	operation-class-aware thermodynamic bridges
SIT79-H-002	purity equality implies unitary dynamics	retracted	channel- and state-specific equality conditions
SIT79-COH-001	the model amplitude is directly local purity	separated	an explicit state-to-field calibration map
SIT79-EM-001	physical electromagnetism is derived from the coherence phase	replaced as a general derivation; Maxwell kept independent	optional global phase or hidden-U(1) models
SIT79-POT-001	the negative-quadratic quartic potential is minimized at zero	retracted; zero is a local maximum for positive parameters	alternative stable potentials and backgrounds
SIT79-MG-001	a scalar lump proves the Yang-Mills mass gap	retracted	the separately governed Yang-Mills and Branch C programs

Correction ID	Version 79 statement	Draft 13 source disposition	What remains open
SIT79-GR-001	assumed standard sectors are derived	retained with boundary as recovery in declared limits	an actual emergence or uniqueness theorem
SIT79-PRED-001	fixed near-term anomalies follow from an unspecified family	retracted as predictions	controlled residual experiments for frozen benchmarks

These corrections do not remove the historical questions: whether temporal structure admits a useful field description, whether coherence-related preparation can couple to that structure, whether gradients can produce residual motion, or whether mismatch dissipation supplies a common computational architecture.

12. Companion-paper architecture and Version 78 recovery

SIT becomes more defensible when each extended bridge has its own methods and failure conditions. One such bridge asks how a system’s history changes its later control response and whether structured dissipation can support conditional agency; the companion program calls that history-conditioned organization teleonomy.

SIT branch	Current governed destination
QGTCD temporal-opportunity model	<i>Quantum Gradient Time Crystal Dilation: Classical Temporal-Opportunity Kernels and a Conditional Newtonian Calibration</i>
time density and local quantum mechanics	<i>Super Dark Time: Gravity Computed from Local Quantum Mechanics</i> revision lane
contractive mismatch dynamics	<i>A Mathematical Framework for Contractive Computational Dynamics with a Thermodynamic Bridge</i>
teleonomy and history-conditioned agency	Paper 75, <i>History-Conditioned Control in Open Systems: Teleonomic Potentials, Dissipative Computation, and Conditional Agency</i> ; governed private Draft 3
SuperTimePosition	<i>SuperTimePosition and Gravitational Phase-Lock</i>
holography / AdS-CFT	<i>SIT, Holography, and the AdS / CFT Translation Split</i>
arithmetic / Riemann	Branch C paper series and theorem ledgers
Causal Fermion Systems and other rivals	governed Comparison Wiki and comparison papers
neuroscience / SAN	SAN papers and SAN Wiki, with no automatic physical-SIT validation

The Version 78-to-79 audit found that the companion-paper goal was incomplete. The largest removed block - the teleonomic framework connecting perturbation, a history-conditioned control landscape, structured dissipation, conditional agency, and empirical perturbation tests - is now converted into governed private Paper 75 Draft 3 with a synthetic-controller benchmark. It remains private and unapproved. That companion does not validate physical SIT. Smaller open blocks concern buoyancy controls, cross-scale fractal / chaos calibration, multiscale renormalization, and a concise conceptual primer. Buoyancy and water / ice / steam remain bounded residual-test and source-custody routes, not current standalone-paper priorities. These subjects remain source material and must not vanish merely because they are outside this paper’s core derivation.

The ordinary-buoyancy explanation in Version 78 must not be restored as established physics. Heating, cooling, freezing, melting, vaporization, and condensation change density, pressure, volume, latent heat, stress, and other standard variables. A future SIT treatment may ask only whether a frozen coupling model predicts an additional equivalence-principle residual after those effects and total mass-energy are controlled.

13. Distinct contribution, inherited tools, and limitations

13.1 The contribution this paper advances

The proposed SIT contribution is the joined research architecture:

1. the documented route from oscillatory coincidence and QGTCD to time density and informational dissipation;
2. the distinction and proposed coupling between event-rate / time-density and coherence-related variables;
3. the weak-field operator that turns a calibrated temporal-field gradient into a conditional residual acceleration;
4. the separation of phase transport from physical electromagnetism while retaining phase as a modeled organizational variable;
5. the cross-paper experiment and falsification program; and
6. the explicit ledger of which bridges are definitions, model consequences, hypotheses, or open companion programs.

That contribution is advanced here as a specific, testable arrangement of operations and source history. The scalar-tensor action with a conformal matter metric, the scalar-dependent Maxwell kinetic term, Yukawa corrections, relative-entropy data processing, and ordinary likelihood methods are inherited scientific tools. The manuscript's claim is not that those ingredients were newly invented. It is that the documented SIT / QGTCD genealogy joined event-rate calibration, coherence-related preparation, a temporal-field gradient, matter response, dissipation, and reversible residual tests into the research architecture formalized here. Whether that conjunction is physically useful remains an empirical and mathematical question; a distinctive narrative does not substitute for a successful model.

13.2 Current limits

Draft 13 does not supply:

- a microscopic derivation of τ or σ ;
- a unique operational kernel in (2) or (4);
- a controlled EFT operator basis, cutoff, power counting, matching, or truncation error;
- a stable cosmological or compact-object solution;
- a full finite-source, multiscale allowed-parameter fit or a screening calculation;
- a quantum theory of the scalar sector;
- a derivation of quantum probabilities, the Standard Model, electromagnetism, or general relativity;
- a proof of the Yang-Mills mass gap, the Riemann Hypothesis, or an SIT / Riemann bridge; or
- an observed SIT-specific anomaly.

These are not reasons to erase the research questions. They are the work required to answer them.

14. Required next calculations

Benchmark A now freezes V, Ω, B_F , identifies the flat-background mass matrix, proves positive linear scalar masses under its stated parameters, solves the point-source force law in the physical matter frame, passes a synthetic paired-fit and uniform-sphere finite-source identifiability benchmark, and supplies an executable likelihood for the published Cassini summary under a declared long-range interpretation. It is a reference calculation, not a viable parameter fit. Before a release candidate, the program should complete the following bounded sequence:

1. replace the published-summary proxy and synthetic paired fit with paired bare-coefficient and dynamically calibrated global fits using massive-body ephemerides, null propagation, and experiment-level nuisance parameters to resolve the disagreement among References 24-26;
2. fold the positive two-mode Yukawa sum through published torsion-balance, geophysical, lunar, planetary, and other relevant response functions to determine the full multiscale allowed region;
3. replace the uniform-sphere demonstration with the actual density profiles and apparatus geometries required by each experiment rather than relying on point sources, one fixed geometry, or a single published contour;
4. derive the full quadratic metric-scalar action and verify gauge-fixed constraint propagation;
5. extend or replace the Minkowski background with one declared cosmological background and test its stability;
6. decide whether a screening mechanism is necessary and, if so, derive it as a new benchmark rather than adding it verbally;
7. define one operational state-to-field kernel and its preparation-energy accounting;
8. generate one preregistered apparatus prediction with uncertainty and null models;
9. complete source-by-source citation and notation review; and
10. obtain independent quantum-information and scalar-tensor review.

If no nonempty region survives, that benchmark should be retired without erasing the earlier genealogy or preventing a different implementation from being tested.

15. Conclusion

The correct repair to Super Information Theory is neither to defend every equation in Version 79 nor to replace the entire program with an eight-page correction note. Draft 13 preserves the valid mathematical corrections, restores the theory's genealogy and coordinating architecture, gives the central QGTCD claim a precise conditional form, exposes the coupling-range tradeoff demanded by existing gravity tests, and turns the Cassini disagreement into an explicit mass-calibration test. It also implements the first bounded paired inverse problem, shows why finite-source geometry can be invisible in residuals while still biasing inferred couplings, and converts the published Cassini estimate into a reproducible sign-constrained summary likelihood without mislabeling it as raw-data analysis. In the baseline model, a time-density or coherence-related field affects motion only by contributing to the physical metric or another declared matter coupling; its gradient then enters the observable potential. That is a derivable model mechanism, not yet an observed law.

SIT's strongest current form is therefore a layered, falsifiable research program. Its historical atoms are preserved by public sources. Its baseline mathematics is explicit. Its errors are recorded. Its broadest bridges are routed to companion papers. Its central empirical burden remains: produce a healthy, constrained benchmark and a residual signal that established physics and preparation accounting do not explain.

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